

Government School Management System (GSMS)

Project Proposal

1. Introduction

Government schools manage numerous academic and administrative activities including admissions, attendance, examinations, teacher management, inventory, library operations, welfare programs and communication. This project proposes a centralized web-based Government School Management System to digitize these processes.

2. Problem Statement

Many government schools rely on manual records or disconnected software, causing delays, errors, duplicate records, poor reporting and inefficient administration.

3. Proposed Solution

Develop a secure web-based platform that centralizes academic and administrative operations with role-based access for administrators, principals, teachers, students, parents and staff.

4. Objectives

Digitize student and teacher management, automate attendance and examinations, manage timetables, inventory, library, welfare programs, communication and reporting.

5. Scope

Modules include Student Management, Teacher Management, Staff Management, Academic Management, Attendance, Timetable, Examination, Library, Inventory, Welfare, Communication, Reports and User Management.

6. Target Users

Principal, Vice Principal, School Administrator, Teachers, Administrative Staff, Librarian, Laboratory Assistant, Students, Parents and Education Officers.

7. Functional Requirements

Student registration, attendance, examinations, report cards, library, inventory, announcements, reports, secure login and role-based access.

8. Non-Functional Requirements

Security, scalability, reliability, responsiveness, usability, backup and maintainability.

9. Technology Stack

Frontend: React.js + Tailwind CSS. Backend: Node.js + Express.js. Database: PostgreSQL. Authentication: JWT + RBAC. APIs: RESTful.

10. Expected Benefits

Reduced paperwork, faster administration, better communication, accurate reporting, improved transparency and better decision-making.

11. Future Enhancements

Mobile app, QR ID cards, AI analytics, automatic timetable generation, face recognition attendance and Ministry integration.

12. Conclusion

The proposed Government School Management System will modernize school administration by providing a centralized digital platform that improves efficiency, transparency and service quality.